

Novel Readout Electronics for Thickness Shear-Mode Liquid Sensors Compensating for Spurious Conductivity and Capacitances

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Abstract—In this paper, a readout circuit for a thickness shear-mode (TSM) resonator is presented which can be used for sensing applications in liquids such as viscosity sensing. The system features compensation of both spurious capacitances and conductances in parallel to the resonator. This allows measurements even in conductive liquids without the need for an elaborate sealing of the sensor. The influence of the spurious elements is determined by means of two orthogonal synchronous detectors and eliminated by active compensation using voltage-controlled amplifiers (VCAs). Furthermore, the circuit is robust against possible phase errors. The basic concept is discussed in detail, and a readout circuit is developed. A prototype is presented, and sample results are given, demonstrating the feasibility of the approach.

Index Terms—Active compensation, capacitance and conductance compensation, electronic readout circuit, microacoustic sensors, thickness shear mode (TSM), viscosity sensors.

I. INTRODUCTION

QUARTZ thickness shear-mode (TSM) resonator sensors (Fig. 1) have been utilized for a number of applications like the measurement of liquid viscosity [1]. The liquid surrounding the sensor changes both its resonance frequency and the damping of the piezoelectric resonator. Recently, several readout concepts have been presented to determine changes in both or one of these parameters. A common approach is the utilization of the TSM resonator as the frequency-determining element in an oscillator circuit [2], [3]. However, with increasing viscosity of the liquid under test, the sensor impedance is dominated by the spurious parallel capacitance of the sensor eventually leading to a malfunction of the oscillator.

To eliminate the influence of the spurious capacitance, several approaches have been suggested, e.g., tuning out the spurious capacitance by means of a parallel inductor [2] or capacitor [4], but the value of the spurious capacitance is not exactly known and is also subject to change [5]. This problem may be circumvented by feeding the resonator by an external oscillator and using a dedicated synchronous detector circuit [6], or an electronic compensation circuit [7].

Liquid conductance disturbs the functioning of the circuit in a similar way. Therefore, it was suggested to establish the contact

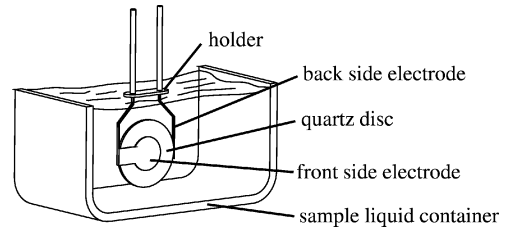


Fig. 1. TSM quartz resonator immersed in a liquid.

with the liquid only on one face of the resonator disk by means of sealing, leading to elaborate packaging.

The presented readout circuit represents a successful implementation of a principle discussed in [8], where also a partial proof of principle is provided. The concept features active compensation circuits for both spurious capacitances and conductances, which is similar to the method used in [7] to compensate the spurious capacitance. In contrast to the approaches mentioned before, this method allows measurements in conductive liquids without the need for device sealing. Compared to other concepts, the readout circuit is thus robust against phase shifts of the amplifiers and the compensation control loops [8].

Below, we discuss the readout concept and present a prototype of a device designed for measuring the viscosity of liquids. Sample measurement results demonstrate the operability of the device.

II. READOUT CONCEPT

The electric equivalent circuit of a TSM resonator immersed in a conductive liquid is shown in Fig. 2 [1]. The motional arm L_1 – C_1 – R_1 represents the mechanical resonance and the internal losses, and C_0 accounts for the static capacitance of the quartz. The acoustic load, i.e., the liquid the sensor is immersed in, is modeled by the additional elements L_2 and R_2 , which depend both on the liquids viscosity density product $\eta\rho$. To first order, L_2 and R_2 are proportional to $\sqrt{\eta\rho}$ [1]. If contact with the liquid is established on both faces of the resonator disk, the finite conductance of the liquid must be considered by an additional conductance G_0 , representing the spurious current between the two electrodes (Fig. 1) through the conductive liquid. The overall sensor admittance is given by

$$Y(j\omega) = \frac{1}{\underbrace{j\omega(L_1 + L_2) + \frac{1}{j\omega C_1} + R_1 + R_2}_{Y_m(j\omega)}} + j\omega C_0 + G_0. \quad (1)$$

The admittance locus is shown in Fig. 3.

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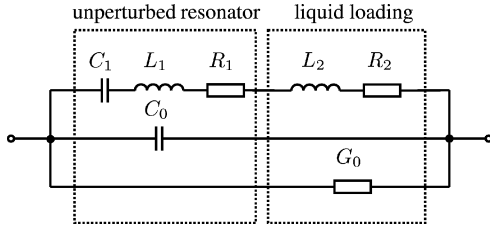
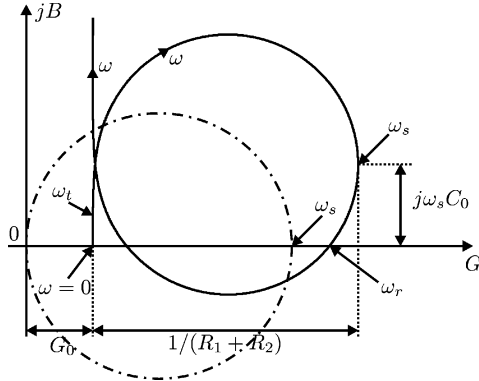


Fig. 2. Lumped element model of the TSM resonator under liquid loading.

Fig. 3. Admittance locus of the TSM resonator under liquid load $Y(j\omega)$ (solid line), and of the motional arm alone ($Y_m(j\omega)$, dotted-dashed line).

Typical values for the equivalent circuit parameters of the 6-MHz AT-cut quartz TSM resonator used are $L_1 = 62$ mH, $C_1 = 11.3$ fF, and $C_0 = 3$ pF. R_1 is in the order of 10 to 100 Ω . These values were determined by measuring the impedance spectrum of the quartz and fitting the equivalent circuit parameters (Fig. 2) to the measured data. The values of L_2 and R_2 depend on the liquid load. The presented readout circuit aims to determine the value of R_2 . It depends strongly on the liquid's properties and can reach values up to several tens of k Ω .

Especially in highly viscous liquids, the motional arm's impedance becomes highly resistive due to the increase of R_2 . The sensor admittance is then dominated by C_0 and G_0 , disturbing the readout of the sensor. To avoid this effect, the presented readout circuit features compensation for both C_0 and G_0 simultaneously. The basic concept of this readout is depicted in Fig. 4.

One terminal of the resonator (represented by its electric equivalent circuit, dashed box in Fig. 4) is connected to an ac voltage source, whereas the second terminal is virtually grounded by the low input resistance of a transimpedance amplifier. The input current I of the amplifier A_1 is thus partly given by the resonator sensor current

$$I_{TSM} = I_m + I_{C_0} + I_{G_0} = I_m + j\omega C_0 V_1 + G_0 V_1. \quad (2)$$

For the compensation of C_0 and G_0 , two additional branches consisting of voltage controlled amplifiers (VCAs) with variable gains g_C and g_G , and a capacitor C_c and conductance G_c , respectively, are introduced and connected in parallel to the TSM resonator, yielding additional contributions to the input current I . Thus, the total output voltage of the transimpedance amplifier is

$$V = R[I_m + j\omega V_1(C_0 + g_C C_c) + V_1(G_0 + g_G G_c)]. \quad (3)$$

If one is able to set the values of g_G and g_C according to

$$G_0 + g_G G_c = 0 \quad (4)$$

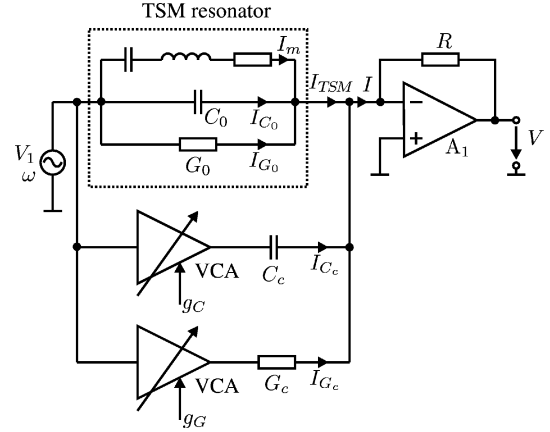


Fig. 4. Circuit for eliminating the spurious capacitance of the TSM resonator and the influence of a conductive liquid.

and

$$C_0 + g_C C_c = 0 \quad (5)$$

the currents I_{C_c} and I_{C_0} , and I_{G_c} and I_{G_0} , respectively, will cancel each other. In this case, V is solely dependent on the current of the motional arm I_m , i.e.,

$$V = R I_m. \quad (6)$$

The conditions in (4) and (5) can be fulfilled by two orthogonal feedback control loops. For this purpose, the circuit in Fig. 4 is driven at a frequency ω_t , that is close but not too close to the series resonance frequency ω_s of the motional arm, i.e., at a frequency where the sensor admittance virtually lies on the vertical line in Fig. 3. In this frequency range, the resonator admittance Y is predominantly determined by C_0 and G_0 , and (3) reduces to

$$V(j\omega_t) \cong R V_1 [(G_0 + g_G G_c) + j\omega_t (C_0 + g_C C_c)]. \quad (7)$$

At this frequency, the in-phase and quadrature components of V correspond to (4) and (5), respectively. Both components are determined by means of an orthogonal demodulator setup. The dc components of the demodulated signals are used as error signals in two feedback control loops, which adjust the VCA gains such that the error signals vanish. As long as the amplifiers do not saturate and behave linearly, this compensation is approximately also valid at the series resonance frequency of the TSM, i.e., ω_s .

If the influences of C_0 and G_0 are fully eliminated, the TSM resonator admittance is determined by $L_1 + L_2$, $R_1 + R_2$, and C_1 . At resonance frequency ω_s , the inductive and capacitive admittances cancel each other, and the total current I is solely determined by $R_1 + R_2$, i.e.,

$$V = \frac{R V_1}{R_1 + R_2} \quad (8)$$

which thus is a measure for the viscosity density product of the liquid under test. For many applications, the liquid density remains nearly constant, while the viscosity changes in a wide range. In these cases, the amplitude of the output voltage V can be considered as a measure for liquid viscosity.

Fig. 5 depicts the block diagram of the complete readout circuit. The circuit accommodates an oscillator running at the

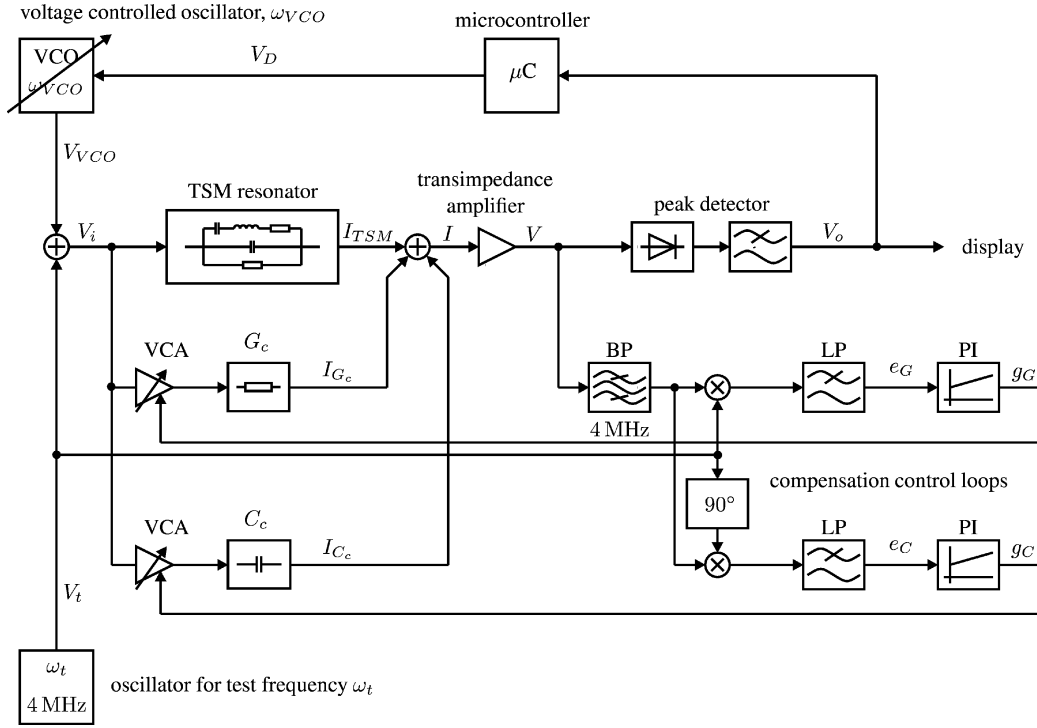


Fig. 5. Block diagram of the readout circuit.

test frequency ω_t (4 MHz), and a second one that ought to be tuned to series resonance frequency ω_s . Thus, the latter oscillator is implemented as a voltage-controlled oscillator (VCO). The output voltages of both oscillators are applied to the resonator using a summing amplifier. As discussed above, the currents from the resonator and the compensating branches are measured by a transimpedance amplifier. The two control loops for the compensation branches maintain the validity of the conditions given in (4) and (5).

The VCO is tuned to the resonator series frequency using a third feedback loop. Provided that compensation of C_0 and G_0 is given, the admittance locus plot for $C_0 = 0$ and $G_0 = 0$ can be considered being equal to the motional arm admittance Y_m (the dotted-dashed line in Fig. 3). Thus, the series resonance frequency ω_s will coincide with the zero phase frequency ω_r . Therefore, tuning the VCO to ω_s could be done by using a phase-locked loop (PLL) feedback system as demonstrated in [7]. As shown in Fig. 3 (the dotted-dashed line), the magnitude of the motional arm admittance Y_m reaches its maximum at ω_s and, hence, the resonator current and the dc output voltage V_o will become maximum at ω_s . This allows the use of a maximum search algorithm as an alternative method for detecting the resonance frequency. In our readout circuit, the latter method was used and implemented in a microcontroller, allowing other modes of operation like the recording of full resonance curves, as well. To obtain a fully analog readout circuit, the PLL approach described before could be used alternatively.

At first sight, one would expect the test signal at ω_t to perturb the measurements at the VCO frequency ω_{VCO} . But in case of full compensation of C_0 and G_0 , the ω_t component of the current I will become zero. Therefore, the transimpedance amplifier (and furthermore the peak detector) will only measure the ω_{VCO} component, which will, provided that ω_{VCO} is tuned to

ω_s , correspond to the viscosity-density product of the investigated liquid.

III. IMPLEMENTATION

A. Circuit Description

The block diagram shown in Fig. 5 is the basis for the realized device. The circuit is split up on two printed circuit boards. The first one accommodates the analog circuitry implementing the complete block diagram except for the microcontroller and the external sensor (TSM resonator). The second board contains the microcontroller, the power supply, the user interface circuits, and the USB interface. It is described in detail in the firmware section.

Two **Colpitts-type oscillators** serve as signal sources using a quartz crystal resonator (determining ω_t), and an $R-L-C$ resonating circuit (ω_{VCO}), respectively. A varicap diode (BBY40) acts as tunable element for the VCO (Fig. 6). The VCO features an output voltage of 0.66 Vrms, the oscillator frequency can be tuned by V_D in the range from 5.97 to 6.01 MHz. The test frequency ω_t is chosen at 4 MHz. Subsequently, a wideband summing operational amplifier (MAX477 300-MHz) sums up the output voltages of the two oscillators.

The **voltage-controlled variable gain amplifiers** (VCA, Fig. 7) are realized by using AD835 250-MHz voltage output four-quadrant multipliers. They feature two differential and one single-ended input and provide a function

$$W = \frac{(X_1 - X_2)(Y_1 - Y_2)}{1 \text{ V}} + Z \quad (9)$$

within an input voltage range from -1 to 1 V [9]. For use as VCA, the pins X_1 , Y_2 , and Z are tied to ground. The input signal is connected to X_2 , whereas Y_1 sets the amplifier's gain. A dc

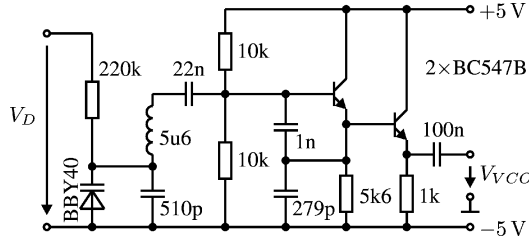


Fig. 6. Colpitts-type voltage controlled oscillator.

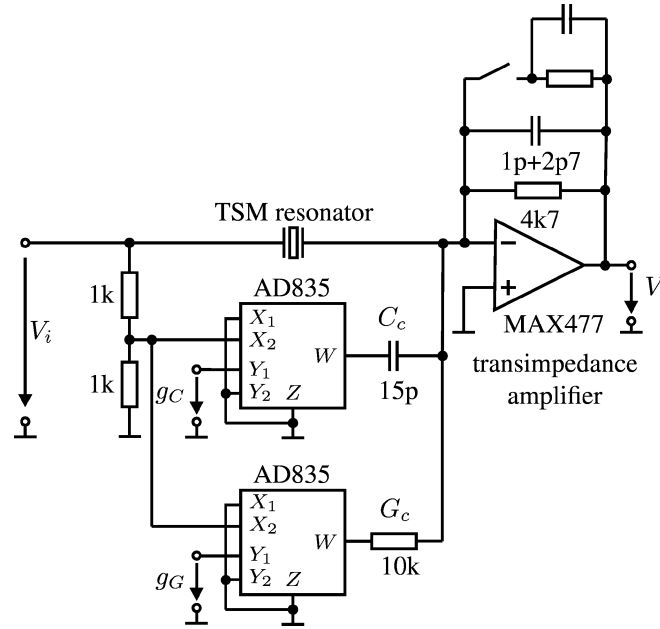


Fig. 7. VCAs using AD835 four-quadrant multipliers and transimpedance amplifier.

voltage of 0.5 V at Y_1 corresponds to an amplifier gain of -0.5 . Note that the output voltage of the summing amplifier is in the range from -2 to 2 V, which would saturate the VCAs. Therefore, a voltage divider with an input output voltage ratio of 2:1 is used. However, this also results in reduced overall compensation branch gains. To compensate this effect, the values of C_c , and G_c have to be increased by a factor of 2. The chosen values of $C_c = 15$ pF, and $G_c = 10^{-4} \Omega^{-1}$ (corresponding to 10 k Ω) allow for compensation of spurious capacitances $C_0 \leq 7.5$ pF, and conductances $G_0 \leq 5 \cdot 10^{-5} \Omega^{-1}$ ($1/G_0 \geq 20$ k Ω), respectively.

The **transimpedance amplifier** circuit (Fig. 7) also uses a MAX477 operational amplifier with a feedback resistor. Additional capacitors prevent the amplifier from spurious oscillations caused by the capacitance of the cable which connects to the TSM resonator. A switch allows changing the transimpedance amplifier gain. Therefore, the device features two different measuring ranges.

The two **synchronous demodulators** for the G_0 and C_0 compensation loops are realized with MC1496 balanced modulator/demodulator integrated circuits. The MC1496 is configured as a differential amplifier, which is switched over between positive and negative gain by V_t (see [6] for a detailed schematic of this configuration). In principle, the demodulators detect only the test signal at ω_t and its odd numbered harmonics, and would

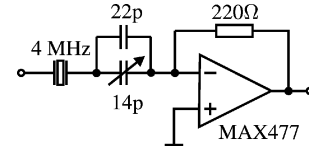


Fig. 8. 4-MHz narrowband filter to prevent synchronous detector input saturation.

therefore suppress the VCO signal at ω_{VCO} . However, to avoid spurious VCO signals in the first place, an additional bandpass filter was used to, e.g., prevent the demodulators from input saturation caused by the VCO signal. This narrowband filter uses a 4-MHz quartz crystal resonator and a MAX477 op-amp (Fig. 8), requiring only a few discrete components. However, due to the large quality factor of the resonator the circuit features a large time constant, leading to overall settling times of a few seconds. For applications like monitoring liquid conditions over a long time span with no abrupt changes in the liquid properties, this does not represent a restriction.

The **low-pass filters** (LP, Fig. 5) at the detector outputs are realized by connecting capacitors to the MC1496's output terminals; the cutoff frequencies are then determined by said capacitors and the output impedances. The remaining dc components of the respective output voltages serve as error signals e_G and e_C in the compensation control loops (see Fig. 5). The time constant of the 4-MHz narrowband filter dominates the transfer function of the control process, which may be approximated by a PT_1 system. The controllers are implemented as PI controllers using uA741 standard op-amps.

B. Controller Firmware

An Atmel ATmega32 microcontroller accomplishes the tuning of the VCO frequency. The internal 10-bit analog-to-digital (A/D) converter digitizes the output voltage V_o of the analog circuit. The controller also steers the VCO by adjusting the varicap diode-bias voltage V_D (Fig. 6) with an external 12-bit digital-to-analog (D/A) converter. From the microcontroller's point of view, the analog circuit is a function $V_o = f(V_D)$ with a maximum at the TSM resonator series resonance frequency [Fig. 9(a), solid line].

For the maximum search algorithm, the controller performs $2n + 1$ measurements at

$$V_{D,k} = V_m + k\Delta V, \quad -n \leq k \leq n \quad (10)$$

where $V_{D,k}$ are the voltages applied to the varicap diode, V_m is the starting voltage, and ΔV is the step size [Fig. 9(a)]. Afterwards, the algorithm searches the maximum value $V_{o,max} = \max_k V_{o,k}$ within the results $V_{o,k}$, i.e., $V_{o,1}$ in Fig. 9(a). It also determines the corresponding VCO voltage $V_{D,max}$. This $V_{D,max}$ forms the new value of V_m , and the measurements start again with decreased step width ΔV [Fig. 9(b)]. After a number of iterations, the current value of $V_{o,max}$ provides a measure for the viscosity-density product of the liquid under test [Fig. 9(c)]. During the measurements, the firmware checks for overflow and compensation errors.

The presented device performs six iterations with step sizes of 1.2, 0.36, 0.18, 0.097, 0.024, and 0.012 V. In the *automatic mode* of the device, the maximum search described above is continuously running. In addition, the firmware provides a *manual*

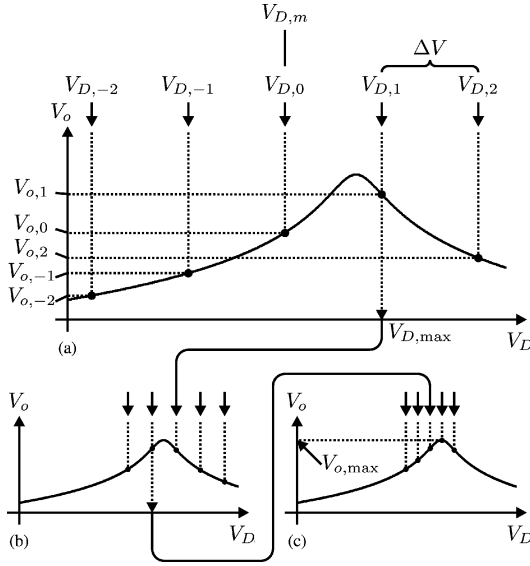


Fig. 9. Algorithm to find the resonance curve maximum for $n = 2$ and three iterations.



Fig. 10. Photo of the readout device.

mode, which allows setting the VCO voltage by the user, and a *remote mode* to transmit complete resonance curves $V_o = f(V_D)$ to a computer via USB interface.

A picture of the realized device is given in Fig. 10.

IV. RESULTS

To demonstrate the operability of the readout system, a series of measurement has been carried out using insulations oils (samples 1 to 6) and silicone oil (AK350) as sample liquids. The insulation oils 1 and 2 are of the same type (Technol US3000), but stem from different batches, whereas the samples 3 to 6 represent aged oils of the same type.

Fig. 11 shows the TSM resonator equivalent circuit parameter $1/(R_1 + R_2)$ versus the readout voltage of the device. The values of $1/(R_1 + R_2)$ were obtained by means of a network analyzer (Hewlett-Packard 8753E). From the measured frequency characteristics, the equivalent circuit parameters have been obtained using a fit algorithm.

Fig. 12 shows the readout of our device in comparison with the sample liquid viscosities, obtained by classical viscosity measurement by means of a cone/plate rheometer (Brookfield

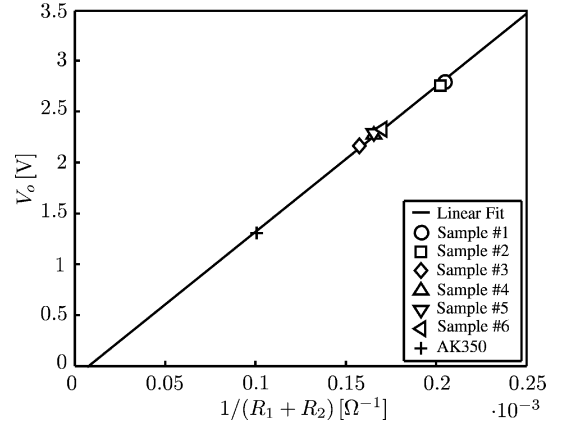


Fig. 11. Motional arm conductance at resonance frequency $1/(R_1 + R_2)$ versus output values of the device presented in this paper.

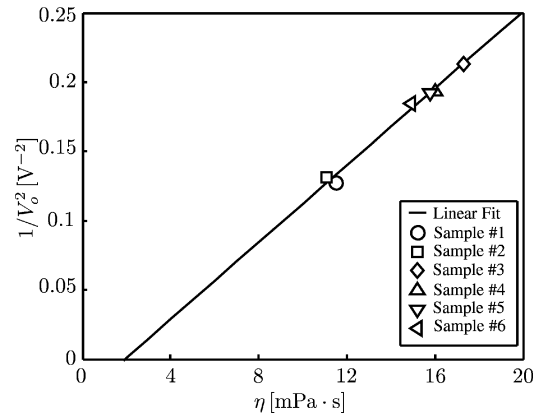


Fig. 12. Sample liquid viscosity versus squared reciprocal output values of our device. The liquid viscosities were measured using a Brookfield LVDV-II+CP cone/plate rotational viscometer.

LV-DV+CP). The plot shows the squared reciprocal readout voltage $1/V_o^2$, which, according to the theory, should be proportional to the viscosity density-product or, in case of virtually identical densities, proportional to the viscosity. Potential causes for deviations from the nearly linear relationship are small differences in the densities of the considered liquids and the non-Newtonian behavior of some liquids.

The measurement results given in Figs. 13 and 14 demonstrate the ability of the readout circuit to compensate for spurious capacitances and conductances. The diagrams show resonance curves $V_o = f(V_D)$ of a TSM resonator immersed in a sample liquid (insulation oil). In order to consider the spurious effects, additional discrete capacitors and resistors, respectively, were connected in parallel to the resonator. The curves were recorded using the readout circuit's remote mode. Fig. 13 shows the results for an additional capacitor C_p (1 pF), whereas a parallel conductance G_p (1/47 kΩ) was used in Fig. 14. For the compensation of significantly larger spurious capacitances or conductances, the values for C_c and G_c have to be increased accordingly (see Section III).

V. CONCLUSION

We have presented a readout circuit for thickness shear-mode quartz resonators. It implements a novel concept allowing for compensation of spurious conductivity and capacitance and is

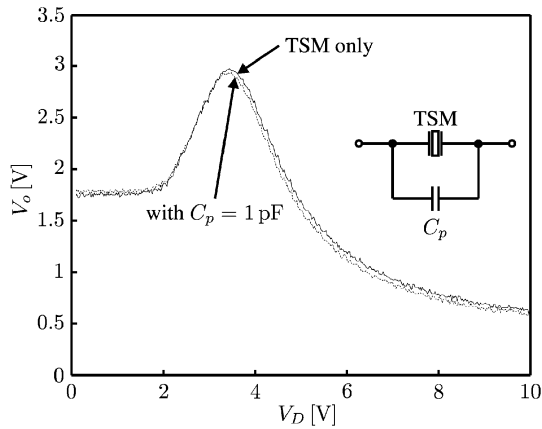


Fig. 13. Measurement of the TSM resonance curve with and without an additional capacitor $C_p = 1$ pF connected in parallel, corresponding to a reactance of ≈ 26.5 k Ω at 6 MHz representing a 37% increase of the resonator's spurious capacitance ($C_0 = 2.7$ pF).

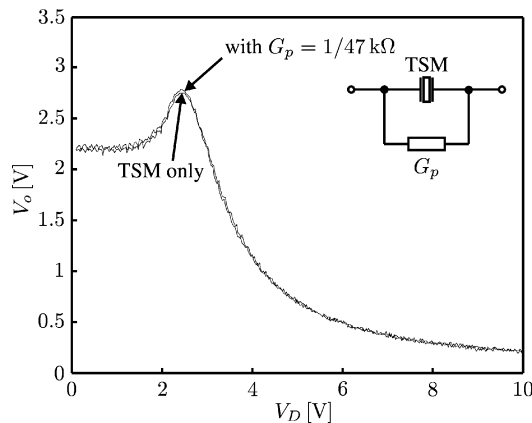


Fig. 14. Measurement of TSM resonator resonance curves with and without a 47-k Ω resistor connected in parallel.

therefore well suited for the measurement of highly viscous and conductive liquids. The electronics aims to determine the additional resonator damping caused by liquid loading and provides a measure for the liquid's viscosity density product. A prototype incorporating the analog readout electronics and a microcontroller has been built to demonstrate the feasibility of the concept.

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